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Enhancing Autonomous Driving Capabilities Using the CARLA Simulator

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Abstract

In this project, an autonomous driving agent was implemented and subsequently improved using the **CARLA simulator**. The starting point was a basic agent with very limited navigation capabilities: it followed a predefined path and stopped in the presence of pedestrians, traffic lights, or other vehicles. However, due to these limitations, the agent often collided with obstacles and was unable to handle situations requiring more complex decisions. To overcome these limitations, it was decided to use exclusively the information provided by CARLA for perceiving the surrounding environment. This data formed the basis for implementing various driving logics capable of handling the given scenarios. The main objective was therefore to enhance the agent by equipping it with advanced features to manage more complex situations, such as interacting with bicycles, overtaking parked vehicles, navigating intersections, managing road cones, and other cases not covered by the initial version. The final result was a **more evolved, robust, and versatile agent**, capable of handling a wide variety of road scenarios safely within the defined scope of the project.

Chapter 1

Introduction

Autonomous driving represents one of the most advanced and complex challenges in the field of artificial intelligence. In this project, we aimed to develop an agent capable of handling complex situations within the CARLA simulator. We started from a basic version of the agent, equipped with essential functions, able only to follow a predefined route and stop in front of visible obstacles, but inadequate for dealing with more dynamic and unforeseen scenarios. Among the complex situations that needed to be addressed, the following were identified:

- **Lane changing:** the agent’s ability to safely switch lanes when required;
- **Intersection handling:** correctly managing intersections by respecting right-of-way rules and deciding when to proceed;
- **Interpretation of traffic lights and road signs:** accurately recognizing and responding to traffic signals and signs, in compliance with road regulations;
- **Interaction with pedestrians, cyclists, and other elements:** safely navigating environments shared with vulnerable road users and other unpredictable obstacles.

Effectively managing these scenarios enabled the development of more advanced driving strategies, leading to a **gradual improvement in the agent’s behavior** and overall performance.

1.1 Methodology

This section outlines the methodology adopted to enhance the autonomous driving capabilities of our agent by leveraging the functionalities offered by the CARLA simulator. Our work focused on addressing a range of complex driving scenarios, with the aim of ensuring that the agent could navigate safely, efficiently, and responsively.

The approach followed was structured into two main phases, each of which represented a fundamental component in the development of the agent:

Identification of Driving Scenarios The first part of our study focuses on the identification of driving scenarios. Through an in-depth analysis of the simulations, we define the critical driving contexts that our agent must be able to recognize and manage. This phase was essential in shaping an agent capable of operating in a highly dynamic and often unpredictable environment. Proper understanding of the scenarios made it possible to anticipate critical behaviors and design consistent and effective responses.

Implementation Strategies Next, we outline the implementation strategies adopted to address the various driving scenarios. This section describes our methodical approach to translating the identified challenges into concrete driving logics, enabling the agent to act effectively and promptly.

1.2 CARLA Controllers

During our work with the CARLA simulator, we employed a series of methods to acquire and interpret the necessary information for the agent's navigation and environment perception. We used the filter method to identify specific types of actors within the simulation, such as vehicles, pedestrians, and other relevant elements. To understand distances and predict potential collisions, we relied on bounding boxes, which allowed us to determine the physical boundaries of objects and their relative positions to the agent. This was crucial for developing obstacle avoidance systems and ensuring safe driving. Additionally, we leveraged waypoints and junctions to guide the agent through the simulation's roads, ensuring it followed logical paths and adhered to traffic rules. Waypoints were particularly useful for defining precise trajectories and managing the agent's behavior near intersections. Lastly, we examined the yaw angles of objects, indicating their orientation in space, and used `get_location()` and `get_transform()` to obtain detailed information about the position and orientation of actors. This data was essential for enabling the agent to fully understand its context and react accordingly. These tools and methods provided us with a comprehensive picture of the simulation environment and equipped the agent with the necessary information to make informed decisions and navigate effectively within CARLA's virtual world. Particular attention was given to the **PID and Stanley controllers**, for which we retained the default parameters provided by the baseline agent. These values were deemed sufficiently appropriate to ensure stable and accurate vehicle control under various test conditions. No significant modifications were introduced, as the observed performance consistently met the requirements of the simulation scenarios, providing a suitable balance between control stability and responsiveness.

PID and Stanley Controller Parameters

```
"pid_control" : {
  "K_P": 0.888,
  "K_I": 0.0768,
  "K_D": 0.05,
  "dt": 0.05
}
"stanley_control" : {
  "K_V": 4.0,
  "K_S": 1.0,
  "dt": 0.05
}
```

Listings 1: PID and Stanley controller parameters used for longitudinal and lateral control.

Chapter 2

Driving Scenarios And Routes

2.1 Driving Scenarios Identified

During the initial phases of the project, several scenarios were identified, related to common and critical situations that the autonomous vehicle must be able to handle promptly, safely, and efficiently. Below is a detailed description of each scenario.

Pedestrian Management: The presence of pedestrians was highlighted, both around curves and behind objects that might not be immediately visible to the agent. In these situations, it is necessary to gradually reduce speed when pedestrians are detected, until a critical threshold is reached, at which point an emergency stop must be performed.

Traffic Lights and Stop Signs: The system must be able to correctly recognize and interpret the color of traffic lights: if the light is **red**, the vehicle must stop; if **green**, it can proceed without interruption. When approaching a stop sign, the vehicle must slow down and come to a complete stop.

Static Objects: If the lane is occupied by static objects, such as cones or warning signs, the vehicle must perform overtaking or lateral offset maneuvers, provided this can be done safely and without fully invading the opposite lane.

Junction Management: Junctions require careful attention to stop signs and different types of intersections. These situations are handled by evaluating the position and movement of surrounding vehicles, merging into traffic only when it is safe to do so.

Vehicles and Bicycles: Scenarios involving parked vehicles along the roadside or bicycles traveling alongside the vehicle or crossing the road unexpectedly have also been considered. In such cases, it is necessary to manage overtaking and lateral movements, as well as apply adaptive cruise control.

The route analysis allowed us to address and manage all the aforementioned scenarios, using information obtained through the tools provided by CARLA, which enabled us to control the actors and determine their positions within the simulated environment.

2.2 Agent Architecture

In the following section, we explain how we implemented the solutions that enable the agent to handle the different driving scenarios introduced earlier. Based on the architecture shown in the previous figure, we created dedicated managers, each designed to manage a specific scenario and its related functions. Below, we provide a detailed description of how each manager works.

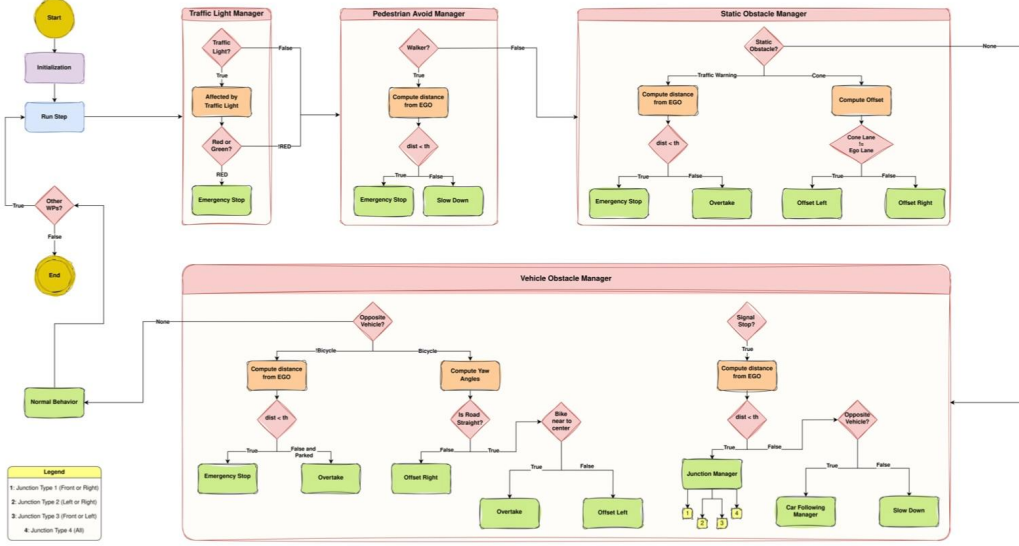


Figure 1: Architecture of our agent’s logic and identified driving scenarios.

2.2.1 Pedestrian Manager

The **Pedestrian Manager** is responsible for ensuring the safe and responsive handling of pedestrians during autonomous driving. Based on Figures 2 and 3, two main risk scenarios were identified: the first involves pedestrians who may suddenly cross the roadway, while the second occurs when pedestrians are near curves where visibility is inherently limited. To effectively address both situations, the system continuously monitors the presence of pedestrians within a defined detection radius. Specifically, a radius of 12 meters is used when the vehicle is approaching or navigating an intersection, whereas a larger radius of 20 meters is employed in all other driving contexts. This configuration ensures that the agent has sufficient time to gradually reduce speed and adapt its behaviour to the specific situation. If a pedestrian enters the roadway—i.e., within the vehicle’s immediate area of interest—and the distance falls below a predefined safety threshold, an emergency braking manoeuvre is automatically triggered to prevent collisions and guarantee maximum safety. To maintain optimal performance even under limited visibility conditions, such as in curves, the vehicle’s front field of view has been widened, allowing the system to detect pedestrians earlier and react more appropriately. If the detected distance remains above the emergency threshold, the vehicle slightly decelerates and continues towards the next waypoint, consistently ensuring a high level of safety and respect for pedestrians and the surrounding environment.

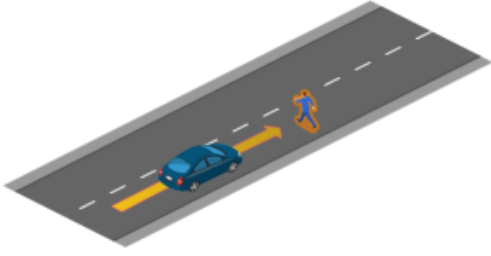


Figure 2: The ego-vehicle encounters a pedestrian on the road and must perform an emergency brake or an avoidance maneuver.



Figure 3: While performing a maneuver, the ego-vehicle encounters an obstacle in the road, either a pedestrian or a bicycle, and must perform an emergency brake or an avoidance maneuver.

2.2.2 Stop Sign Manager

The **Stop Sign Manager** is responsible for ensuring the vehicle's correct and timely response to stop signs encountered along the route, as illustrated in Figure 4. The underlying logic follows a methodology similar to that adopted for pedestrian management: the system detects a stop sign at a distance of 20 meters, providing the agent with ample time to adapt its speed accordingly. Once the stop sign is detected, the system continuously monitors the decreasing distance to the sign and progressively reduces the vehicle's speed as it approaches. If the detected distance falls below a predefined threshold of 2 meters, indicating immediate proximity to the stop sign, the vehicle performs an emergency stop to fully comply with traffic regulations and ensure road safety. Conversely, if the distance remains above the threshold, the vehicle decelerates gradually and continues its trajectory towards the next waypoint, maintaining a smooth and controlled driving behavior. Additionally, the system performs a contextual check in situations where there are vehicles ahead. If stationary vehicles are detected, the system verifies whether they are stopped due to a stop sign, a traffic light, or another obstacle. This verification is essential to prevent the agent from attempting an inappropriate overtaking maneuver when facing a stationary vehicle. By understanding the reason for the halt, the agent can adapt its response, thus avoiding risky overtaking actions. The same precautionary strategy is applied when the vehicle approaches an intersection: in these scenarios, overtaking is explicitly avoided to prioritize safety and compliance with traffic norms.



Figure 4: The ego-vehicle encounters a pedestrian on the road and must perform an emergency brake or an avoidance maneuver.

2.2.3 Traffic Light Manager

The **Traffic Light Manager** is responsible for safely and consistently managing the vehicle’s interaction with traffic lights encountered along its route. Similar to other control modules, all traffic lights are detected at a distance of 50 meters, enabling the system to anticipate and effectively handle potential urban scenarios with heavy traffic near signalized intersections. As the vehicle approaches a traffic light, the system continuously monitors and analyzes its current state in real time. If the traffic light is red, a controlled braking maneuver—or, if necessary, an emergency stop—is initiated to ensure the vehicle halts safely and in full compliance with traffic regulations. Conversely, if the traffic light is green, the vehicle maintains its planned speed and proceeds smoothly to the next waypoint without unnecessary interruptions, thereby preserving driving fluidity and efficiency. This strategy ensures that the agent adheres strictly to traffic rules while exhibiting safe and predictable behavior, minimizing the risk of collisions and enhancing its interaction with other road users. Such an approach is particularly beneficial in complex urban environments characterized by high traffic density and frequent stop-and-go situations.

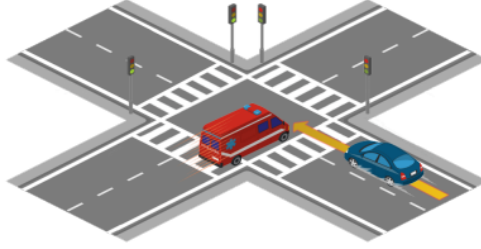


Figure 5: Ego-Vehicle loses control due to bad conditions.

2.2.4 Static Obstacle Manager

The **Static Obstacle Manager** is responsible for the detection and safe handling of fixed obstacles along the vehicle’s route, as illustrated in Figures 6 and 7. Such obstacles include temporary road signs, safety barriers, and traffic cones typically used to demarcate construction zones or lane deviations. Obstacles are classified and managed differently depending on whether they completely block the driving lane, partially occupy it, or partially intrude into the opposite lane.

Overtake When an obstacle fully obstructs the lane, the agent initiates an overtaking manoeuvre by dynamically generating a new sequence of waypoints to follow. To ensure the manoeuvre is performed safely and efficiently, a comprehensive logic has been developed that takes into account multiple contextual factors, including road geometry, the autonomous vehicle’s speed, and the presence of additional obstacles or other vehicles along the route. The key components of this approach are described below:

- **Overtake Distance Calculation** First, the total distance the vehicle must travel to safely complete the overtaking manoeuvre is calculated. To this end, the Pythagorean theorem is employed to approximate the diagonal distance required for a lane change, modelling the lane width and vehicle length as the two legs of

a right triangle, with the hypotenuse representing the actual displacement. The total overtaking distance $d_{overtake}$ is defined as:

$$d_{overtake} = d_{obstacle} + d_{same} + 2 \times hypotenuse + d_{other}$$

where $d_{obstacle}$ denotes the distance from the obstacle to be overtaken, d_{same} indicates the distance travelled in the same lane before initiating the lane change, and d_{other} accounts for the additional distance travelled in the overtaking lane, dynamically computed based on the spacing between parked vehicles encountered beyond the first obstacle (refer to the implementation code for further details). This dynamic calculation ensures that the overtaking distance adapts to the actual distribution of vehicles, providing a flexible and effective manoeuvre.

- **Overtake Time Estimation** Once the overtaking distance is determined, the time required to complete the manoeuvre is estimated using the uniformly accelerated motion equation:

$$OvertakeTime = \frac{-v_0 + \sqrt{v_0^2 + 2as}}{a}$$

where v_0 is the vehicle's initial speed, a is the acceleration, and s is the overtaking distance. This estimate allows the system to verify whether the manoeuvre can be executed safely within the available time and road conditions.

- **Collision Checking** To ensure safety during the overtaking process, collision checks are performed to detect potential conflicts with oncoming traffic. It is assumed that vehicles in the opposite lane travel at the designated speed limit, which allows for their future positions to be predicted under the constant speed hypothesis. The relative distance to these vehicles is calculated within a defined look-ahead interval along the vehicle's trajectory. If a potential collision is detected during the manoeuvre, it is immediately cancelled to prevent accidents. Provided no collision risks are identified and road conditions are favourable, an optimised path is generated that includes waypoints for the lane change, the overtaking phase, and a safe return to the original lane.



Figure 6: The ego-vehicle encounters an obstacle blocking the lane and must perform a lane change into traffic to avoid it. The obstacle may be a construction site, an accident or a parked vehicle

Lateral Offset In scenarios where the lane is partially narrowed—such as when cones are positioned without fully blocking the lane—the agent handles the situation by applying a controlled *lateral offset*, which may be positive (+0.2) or negative (-2), depending on whether the obstacle is located in the ego lane or the opposing lane. This strategy ensures smooth traffic flow while maintaining vehicle safety without requiring a full overtaking manoeuvre. Once the restricted area is cleared, the lateral offset is reset to its nominal value of zero, restoring the vehicle’s original alignment within the lane.



Figure 7: The ego-vehicle encounters an oncoming vehicles invading its lane on a bend due to an obstacle. It must brake or maneuver to the side of the road to navigate past the oncoming traffic.

2.2.5 Junction Manager

The **Junction Manager** plays a crucial role in ensuring that the autonomous vehicle navigates intersections safely, efficiently, and in compliance with standard traffic behavior. Once the presence of an intersection is detected nearby, the system initiates a classification process to identify its configuration among four main types:

- **Type 1:** The intersection allows the vehicle to go straight or turn left.
- **Type 2:** The intersection allows the vehicle to go straight or turn right.
- **Type 3:** The intersection allows left and right turns only.
- **Type 4:** The intersection allows movement in all directions.

Once the intersection is detected and classified, the Junction Manager plans the appropriate maneuver based on the road layout and the vehicle’s intended direction. Leveraging CARLA’s traffic management capabilities, which ensure that only one vehicle enters the intersection at a time, the system significantly reduces the likelihood of collisions. The decision-making logic incorporates a mechanism for monitoring surrounding vehicles by analyzing their active turn indicators at the exit waypoints. This feature exploits the common driving convention of signaling before turning, providing a reliable cue of other vehicles’ intended trajectories. By continuously monitoring the presence and status of these indicators, the Junction Manager gains valuable information about the intentions of other road users, thereby enhancing trajectory planning and minimizing the risk of conflicts.

For each scenario, the Junction Manager applies a structured and adaptive strategy tailored to the specific intersection type:

- **Type 1 (Straight and Left):** In this configuration, the vehicle may proceed straight or turn left. The manager carefully analyzes the presence and speed of vehicles ahead and to the left, determining the safest moment to cross or turn, prioritizing both traffic flow continuity and safety.
- **Type 2 (Straight and Right):** When the intersection permits going straight or turning right, the system monitors vehicles ahead and on the right. If the path is clear, the vehicle proceeds smoothly; otherwise, it yields appropriately to comply with traffic regulations.
- **Type 3 (Left and Right):** For intersections that allow left and right turns only, the manager assesses the vehicles in the left and right lanes, adjusting the vehicle's speed and trajectory to avoid conflicts and maintain a safe crossing, even in dense traffic.
- **Type 4 (All Directions):** Type 4 intersections, which allow unrestricted movement in all directions, represent the most complex scenario and require comprehensive trajectory analysis. In such cases, the manager coordinates maneuvers by simultaneously considering vehicles approaching from all directions, modulating progress according to detected conflicts and adopting a cautious behavior when necessary.

This modular and adaptive approach ensures that intersection traversal is performed in a controlled manner, minimizing unnecessary delays while guaranteeing compliance with traffic rules and maximizing overall road safety.



Figure 8: The ego-vehicle needs to negotiate with other vehicles to cross. This scenario occurs at both signalized and non-signalized junctions.

2.2.6 Vehicle Manager

The **Vehicle Manager** is responsible for the dynamic handling of both motor vehicles and bicycles encountered along the driving route. When the autonomous vehicle is following another vehicle that is moving steadily within the traffic flow, the agent engages an *adaptive cruise control* mode, maintaining a safe following distance and smoothly adjusting its speed in response to the leading vehicle's behavior. A dedicated mechanism has also been implemented to assess the status of the preceding vehicle, distinguishing whether it is parked, stopped at a stop sign, or halted at a traffic light. This capability to accurately interpret the reason for the vehicle's stop enables the system to select the most appropriate response: if the vehicle ahead is parked or obstructing the lane unexpectedly, the agent safely initiates an overtaking maneuver. Conversely, if the stop is due to legitimate traffic control, the autonomous vehicle remains stationary at a safe distance. As an additional safety measure, overtaking is never performed when approaching or navigating a curve, thereby minimizing the risks associated with limited visibility and constrained maneuvering space. Regarding bicycles, the decision-making logic adopts principles similar to pedestrian handling, while incorporating additional caution due to bicycles' higher speeds. When a bicycle is detected within a defined safety range, the agent reduces its speed accordingly to maintain an appropriate buffer. If the bicycle is positioned on the right edge of the lane, the vehicle performs an overtaking maneuver while maintaining a lateral offset of approximately 2 meters, provided there are no curves nearby. If the bicycle occupies the center of the lane, overtaking is executed only when the road segment is sufficiently straight and visibility is clear, in order to avoid hazardous maneuvers on curved or winding sections. The presence of curves is accurately detected by analyzing the spatial coordinates and angular changes of upcoming waypoints: significant variations in direction or heading indicate a curve ahead, triggering a more cautious strategy and restricting overtaking operations. Through this modular and adaptive approach, the Vehicle Manager ensures that the autonomous vehicle maintains compliant, safe, and considerate behavior towards all road users, supporting smooth and reliable navigation in diverse traffic scenarios.



Figure 9: A car is parked on the side of the road, partially blocking the lane. The ego-vehicle must overtake it.

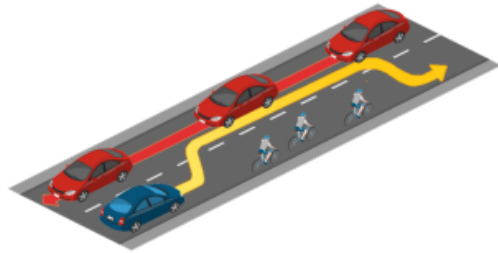


Figure 10: The ego-vehicle encounters a slow moving hazard blocking part of the lane. The ego-vehicle must brake or maneuver to avoid it next to a lane of traffic moving in the opposite direction.

2.3 Routes

The various managers described in the previous paragraph are used to ensure that the vehicle responds appropriately in all the previously identified scenarios. These managers are essential components for guaranteeing the correct behavior of the vehicle during tests conducted on two specific routes, which will be described in the following paragraphs.

2.3.1 Route 1

Route 1 takes place in a nighttime setting, further complicated by the possibility of rain along the route. These conditions make the driving environment particularly challenging due to reduced visibility and reflective surfaces.

Base Agent: The base agent showed significant limitations in this scenario. The simulation ended in failure due to a timeout and a collision with a fixed traffic sign. Although the agent has basic environmental perception, it exhibited serious difficulties in planning and executing driving tasks under nighttime and rainy conditions. Its performance was noticeably worse compared to what was observed in more favorable, daytime scenarios.

Improved Agent: The Base Agent, initially tested at a speed of around 40 km/h, revealed several critical issues, particularly related to collisions and overall penalty scores. To better understand the agent's behavior and improve its performance, a series of experiments was conducted at lower speeds, in order to observe its behavior under more conservative conditions. This approach made it possible to analyze how driving quality changes with speed, offering valuable insights for applying progressive improvements aimed at finding a better balance between safety, smoothness, and rule compliance.

- **Experiment 1 - Pedestrian Collision (average speed: 30 km/h):** In the first test, the agent successfully completed the entire route (100%) but achieved a relatively low driving **score of 41.89**. A collision with a pedestrian was recorded, which, in the context of this project, is considered the least severe infraction. Several penalties were also recorded for driving too slowly, with speeds ranging between 79% and 89% of the surrounding traffic's average. No other significant infractions occurred: there were no lane departures, signal violations, or vehicle blocks. The total time to complete the route was 488.7 seconds.
- **Experiment 2 - Vehicle Collision (average speed: 31.5 km/h):** Following the results of the first experiment, a second test was conducted by slightly increasing the agent's speed to an average of 31.5 km/h, with the aim of reducing penalties related to low speed. Again, the agent completed 100% of the route and achieved a higher driving **score of 53.49**. However, a collision with another vehicle occurred, which, in the project, is considered one of the most heavily penalized infractions, second only to certain severe rule violations. Penalties for low speed were less significant: the agent maintained speeds between 78% and 98% of

the surrounding traffic, showing smoother driving behavior. The total completion time also improved, dropping to 478.95 seconds.

Although the second experiment involved a more penalized type of collision, the slightly increased speed allowed the agent to perform better overall. This suggests that careful speed regulation can lead to more balanced performance, as long as a strong focus on safety is maintained.



Figure 11: *Comparison between the base agent and our agent: **Base Agent** (left) and **Our Agent** (right).* In the first case, the roadwork sign is not detected, resulting in a collision. In the second case, the same static obstacle is correctly identified, and the agent comes to a stop, waiting for the appropriate conditions to safely overtake it.



Figure 12: **Our Agent behavior:** Detection of a parked vehicle in a night-time scenario, followed by evaluation of the adjacent lane to perform a safe overtaking maneuver.



Figure 13: **Our Agent behavior:** Speed adjustment to safely navigate the turn while avoiding collisions with vehicles in the adjacent lane.

2.3.2 Route 4

Route 4 is less complex compared to Route 1, as it takes place in a daytime setting and is free from adverse weather conditions that could compromise the agent’s ability to respond appropriately to the identified scenarios.

Base Agent: Although the base agent was able to complete the entire route, its final score was very low, showing poor performance. It had serious problems interacting safely with the dynamic environment, often crashing into pedestrians and other vehicles. It also ignored stop signs several times. A partial scenario timeout suggests it also struggled with certain situations. Overall, these issues show that the base agent cannot guarantee a safe and compliant behavior in real traffic conditions.

Improved Agent: To make the Base Agent perform better on Route 4, lower speeds were tested instead of the original 40 km/h, which had caused several collisions and penalties. Driving more carefully worked well only at 30 km/h, while at 31.5 km/h there were no clear improvements.

- **Experiment 1 - Collision with static Object (average speed: 30 km/h):** In this experiment, conducted on Route 4 with an average speed of 30 km/h, the agent successfully completed the entire route (100%) and achieved a very good driving **score of 64.23**. The only relevant infraction was a collision with a static object (a stop sign), categorized as a layout collision. A minor penalty was also recorded for slightly low speed, with the agent driving at 96.06% of the surrounding traffic’s speed on average. No other issues occurred: there were no collisions with pedestrians or vehicles, and no violations of traffic lights, stop signs, or lane deviations. The agent completed the route, which was about 900 meters long, in 212 seconds.
- **Experiment 2 - No significant change (average speed: 31.5 km/h):** Following the strong performance at 30 km/h, another test was carried out with the agent’s speed increased to 31.5 km/h. However, **the results showed no significant improvement** over the previous experiment, neither in score nor in driving behavior. For this reason, the experiment was not analyzed or documented in detail, as it did not provide any useful insights for system optimization.

The experiment on Route 4 at 30 km/h confirmed that a moderate speed can lead to very solid performance, with a good score and very few infractions. The subsequent test at 31.5 km/h, having shown no advantages, supports the idea that, at least for this scenario, 30 km/h is the optimal configuration.



Figure 14: *Comparison between the base agent and our agent: **Base Agent** (left) and **Our Agent** (right).* In the first case, the agent fails to detect the vehicle and is unable to perform the overtaking maneuver, while in the second case, correct detection and an effective evaluation of the overtaking are observed.



Figure 15: *Comparison between the base agent and our agent: **Base Agent** (left) and **Our Agent** (right).* In the first case, the child crossing the street is not detected, resulting in a collision. In the second case, the pedestrian is correctly detected, and the agent slows down to allow safe passage.



Figure 16: **Our Agent behavior:** Cyclist detection and corresponding speed adjustment to allow safe passing.



Figure 17: **Our Agent behavior:** Stop sign detection and corresponding wait before resuming movement.

Chapter 3

Results

By returning to the original speed of the Base Agent, **set at 40 km/h**, and integrating an advanced mechanism to dynamically maintain a safe distance from all surrounding vehicles, **the system achieved the best performance recorded so far**. These improvements, along with enhanced robustness in traffic sign recognition under varying environmental conditions, led to near-flawless behavior across both test scenarios, with performance scores consistently close to 100%. These results confirm the effectiveness of the updated control strategies and perception modules in safely and reliably handling complex driving tasks.

3.1 Results of Route 1

On Route 1, the agent demonstrated exceptional performance, completing 100% of the route and achieving an almost perfect composed score of 99.02. This near-flawless result was marked by a total absence of collisions with road layout elements, pedestrians, or other vehicles, as well as full compliance with stop signs and red traffic lights.

There were no deviations from the planned path, no lane departures, no vehicle blockages, and no scenario or route timeouts. The only minor issue observed was a few slight violations for failing to maintain the minimum required speed relative to surrounding traffic, indicating that the agent occasionally drove at a slightly suboptimal speed.

Overall, the outcome on Route 1 reflects a very high level of autonomy, safety, and compliance with traffic regulations.

Results of RouteScenario_1 (repetition 0) — FAILURE —		
Start Time	2025-06-12 16:55:51	
End Time	2025-06-12 17:14:40	
Duration (System Time)	1129.15s	
Duration (Game Time)	440.45s	
Ratio (System Time / Game Time)	0.39	

Criterion	Result	Value
RouteCompletionTest	SUCCESS	100 %
OutsideRouteLanesTest	SUCCESS	0 %
CollisionTest	SUCCESS	0 times
RunningRedLightTest	SUCCESS	0 times
RunningStopTest	SUCCESS	0 times
MinSpeedTest	FAILURE	98.46 %
InRouteTest	SUCCESS	
AgentBlockedTest	SUCCESS	
ScenarioTimeoutTest	SUCCESS	0 times
Timeout	SUCCESS	

Figure 18: Performance and results achieved on Route 1 by the Improved Agent

3.2 Results of Route 4

On Route 4, the agent demonstrated a very high level of performance, successfully completing the entire route with an excellent composed score of 97.28. This execution was nearly flawless, marked by a total absence of collisions with road layout elements (including stop signs—an improvement compared to previous simulations on the same route), pedestrians, or other vehicles.

There were no traffic light violations, significant route deviations, vehicle blockages, or timeouts. The only minor issues observed included a slight and brief lane departure (approximately 15 meters) and occasional violations for failing to maintain the minimum speed required relative to the surrounding traffic flow.

Overall, the performance on Route 4 highlights an extremely high degree of autonomy, safety, and compliance with traffic regulations, representing an almost perfect result.

Results of RouteScenario_4 (repetition 0)			FAILURE
Start Time	2025-06-14 11:24:46		
End Time	2025-06-14 11:33:07		
Duration (System Time)	501.32s		
Duration (Game Time)	225.85s		
Ratio (System Time / Game Time)	0.451		

Criterion	Result	Value
RouteCompletionTest	SUCCESS	100 %
OutsideRouteLanesTest	FAILURE	1.68 %
CollisionTest	SUCCESS	0 times
RunningRedLightTest	SUCCESS	0 times
RunningStopTest	SUCCESS	0 times
MinSpeedTest	SUCCESS	106.84 %
InRouteTest	SUCCESS	
AgentBlockedTest	SUCCESS	
ScenarioTimeoutTest	SUCCESS	0 times
Timeout	SUCCESS	

Figure 19: Performance and results achieved on Route 4 by the Improved Agent